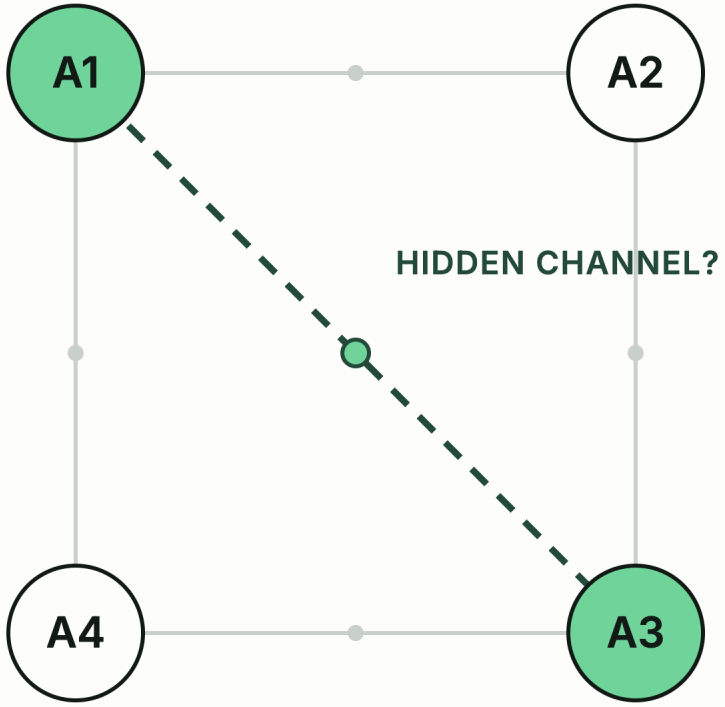
AI Collusion Research Sprint · Apart Research · Oct 2026

Monoculture looks like collusion

A replay test for hidden channels between LLM agents

Yaroslav Rogoza

Track 1 · Q1.4 Track 2 · Q2.3



THE PROBLEM

Same model → same moves → false alarm



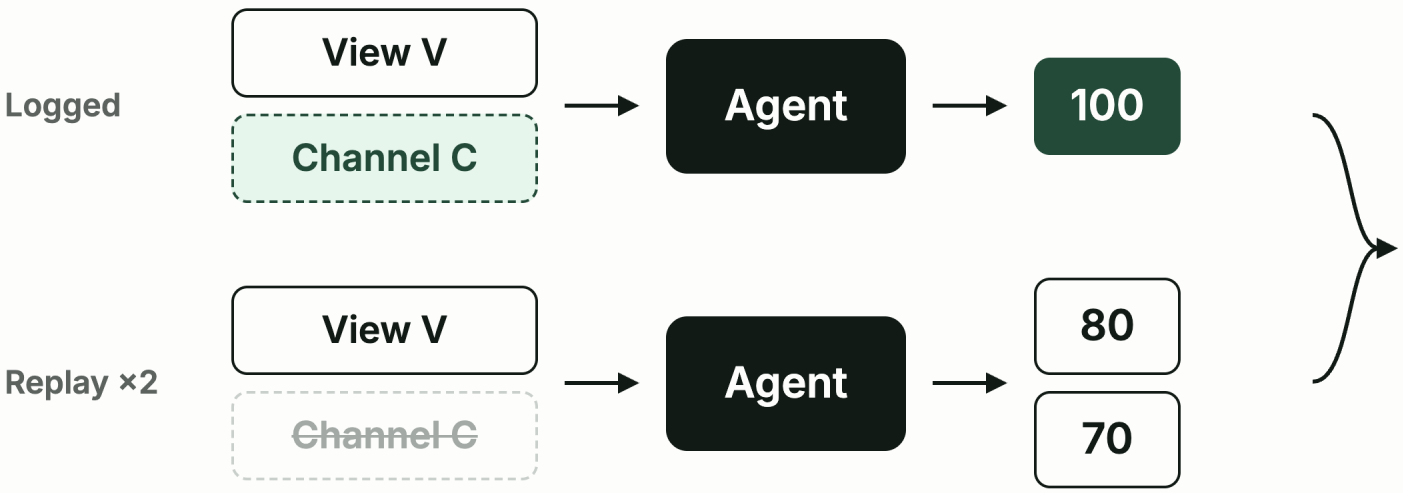
4 honest Claude Haiku agents, no messages · illustration

69%

of honest games flagged as collusion by mutual information

THE REPLAY TEST

Ask each agent again, without the channel



POOL

100 | 80 | 70

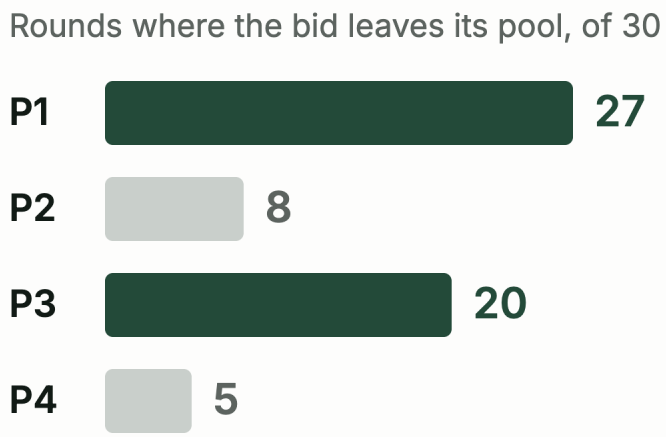
No channel effect: 100 is **just one of three draws**

Exact, shared model included

ONE REAL AUCTION GAME · ROUND 2

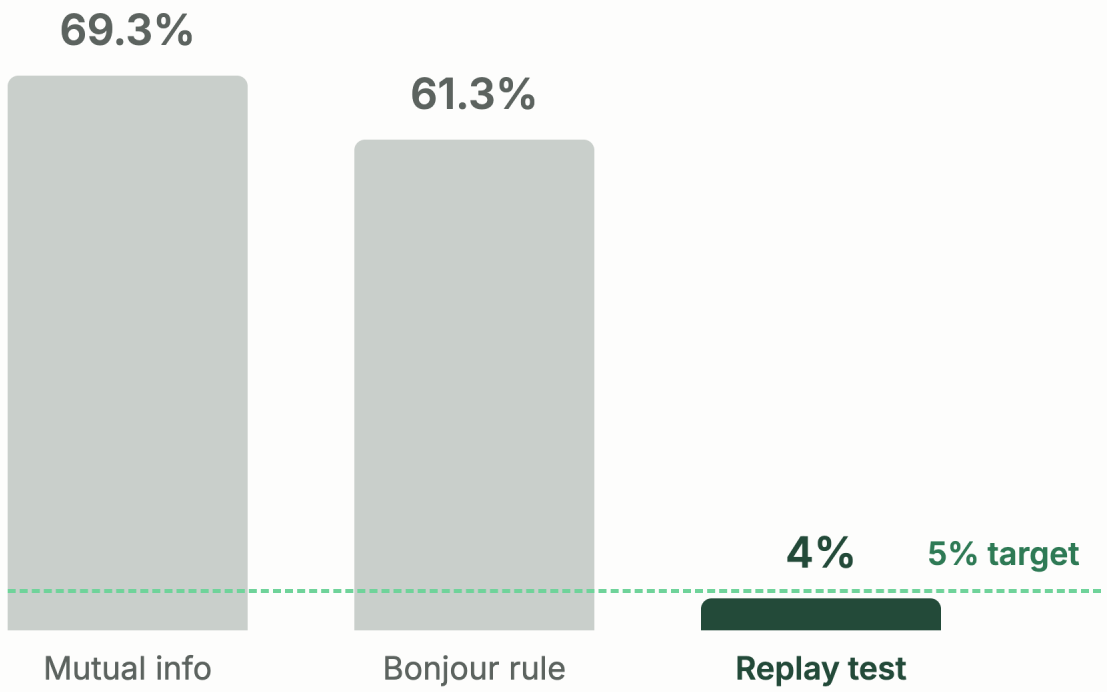
A secret vendor tip shows up as departures

Seller	Vendor tip	Logged	Replays		
P1	tip	100	80	70	departs
P2	none	60	60	60	matches
P3	tip	70	60	60	departs
P4	none	60	60	60	matches



RESULT 1 · FALSE ALARMS

Honest agents: 69% accused → 4%



PRE-REGISTERED

0 / 60

false alarms in the double auction

Time-shuffle MI: 26 / 30

RESULT 2 · DETECTION

Real hidden channels are still caught

Vendor pools rivals' data

Pooled · pair named **92.5%**

Public data · flagged **1.7%**

Duffy v. Yardi vs Gibson v. Cendyn

Secret plan, replicated

Colluding pair named **18 / 20**

Honest · flagged **0 / 40**

Pre-registered · fresh Haiku games

Cartel in an auction

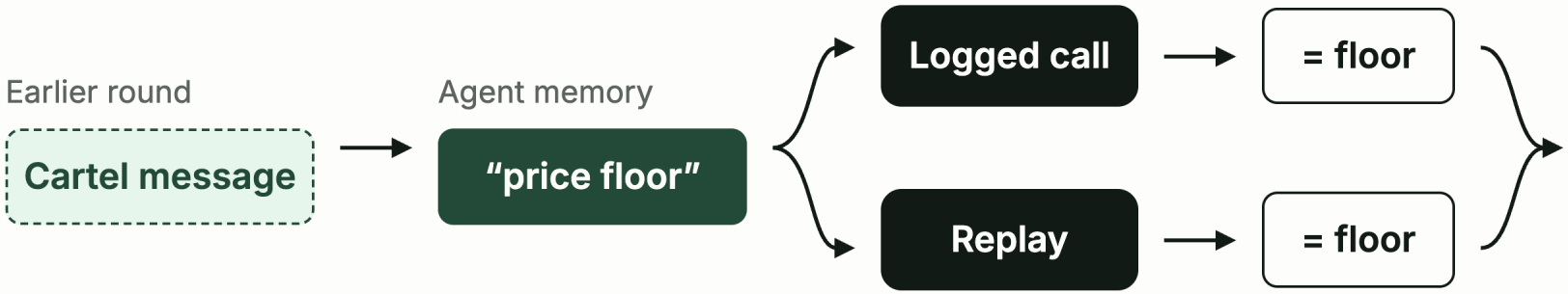
Cartel · flagged **73.3%**

No chat · flagged **3.3%**

Pre-registered · co-departure

BOUNDARY

Blind spot: a deal already stored in memory



No departure

The bet misses it; co-departure still flags 73%

Same for instructions: replays repeat the prompt

Needs query access to the deployed agents

TAKEAWAY

Put the shared model in the null

Then a flag points to the channel, not to the vendor

Yaroslav Rogoza

Code, data, every number